

Lecture 25 Notes: Financing Decisions, Capital Structure, and Financial Markets

Corporate Finance

I. Major Financial Decisions

There are two major types of decisions in finance:

Financing Decisions	Investment Decisions
How to raise capital	Where to invest capital
Debt vs. equity mix	Which projects to pursue
Flexible and reversible	Finality; costly to reverse
Retire bonds, issue new equity	Expand production capacity

Key Insight

Financial markets are highly competitive and dominated by wise investors. Securities are generally **fairly priced**. If a security is not fairly priced, efficient markets will quickly return it to equilibrium.

II. The Pecking Order Theory

Definition

The Pecking Order Theory describes the sequence a firm follows when raising capital, prioritizing sources based on information asymmetry and cost.

The Pecking Order

Internally Generated Funds → Debt → New Equity

1. Internally Generated Funds (First Choice)

- **Retained profits** (earnings not paid out as dividends)
- No flotation costs
- No information asymmetry
- No dilution of control

2. Debt (Second Choice)

- Interest is tax-deductible (tax shield)
- Less information asymmetry than equity
- No dilution of ownership
- Creditors have priority claims

3. New Equity (Last Resort)

- High flotation costs
- Sends negative signal (managers may think stock is overvalued)
- Dilutes existing shareholders' control

III. Limits on Debt Usage

While debt is cheaper and provides a tax shield, companies cannot use an unlimited amount.

Why Debt Increases Cost

1. As a company uses more debt, the **marginal cost of debt increases**
 - Moving from 20% to 40% debt results in higher interest rates
 - Lenders demand higher returns because the firm becomes riskier
2. High debt levels transfer risk to **equity holders**
 - Shareholders worry that interest costs will eat up profits
 - They raise their **required return**
3. This increases the **WACC**
 - Higher WACC can turn project NPV negative
 - Makes it difficult to find profitable investments

Key Insight

There is an optimal capital structure where the tax benefits of debt are balanced against the costs of financial distress.

IV. Book Value vs. Market Value

Comparison

Component	Book Value	Market Value
Bank Loans	Little difference	Little difference
Bonds	Historical cost	Changes with interest rates
Equity	Historical cost	Shares $\times$ Current market price

Market Value of Equity

$$\text{Market Value of Equity} = \text{Shares Outstanding} \times \text{Current Stock Price}$$

Key Observations

- Market value of equity is usually **much higher** than book value
- Market crashes can bring stock prices down near book value
- Using market value weights makes a company appear **less risky** because high equity value lowers the perceived debt ratio

V. The 2008 Financial Crisis: A Cautionary Tale

Background

High debt ratios become particularly dangerous during a recession, leading to financial distress.

Mortgage-Backed Securities (MBS) and CDOs

- Instead of waiting 20-25 years for homeowners to repay loans, banks pooled mortgages
- They issued securities against these pools to get immediate cash from investors
- These securities were divided into **tranches** (senior and junior layers)

The Collapse

- When the housing market crashed, people defaulted on mortgages
- The underlying cash flows for these securities disappeared
- Values of MBS and CDOs plummeted
- Major institutions collapsed: **Lehman Brothers, Bear Stearns**

Lesson

Excessive leverage and complex financial products can lead to systemic collapse when underlying assets deteriorate.

VI. Ownership, Governance, and Control

Major Players in Stock Markets

- Household sector
- Mutual funds
- Pension funds
- Institutional investors (often take active role in management)

Board of Directors

- Represent shareholders
- Elected through direct voting or **proxies**
- Proxies allow influential groups to represent smaller investors

Classes of Stock (Example: Google)

Share Class	Voting Rights	Cash Flow Rights
Founders' Shares	10 votes per share	Same dividends
Ordinary Shares	1 vote per share	Same dividends

Key Insight

Different share classes allow founders to maintain control while raising capital from public investors, even though cash flow rights (dividends) remain equal.

VII. International Debt and Bonds

Eurobonds

- Bond denominated in a currency different from the country where it is issued
- Example: A U.S. dollar bond issued in Pakistan
- Pakistan has issued Eurobonds

Panda Bonds

- Bonds issued in the Chinese market
- Pakistan is currently looking to issue Panda bonds

Issuance Methods

Method	Description
Public Placement	Bonds are sold to the general public
Private Placement	Bonds are sold directly to major buyers (banks, insurance companies, mutual funds)

VIII. Venture Capital and Startups

The Startup Journey

- Many multi-billion dollar companies started as small projects
- **Facebook:** Began as a college project in a Harvard dorm
- **Apple, Google:** Started small
- The most important asset for a new firm is often a **good idea**

Venture Capital (VC)

Definition

Venture Capital (VC) provides equity capital to young, disruptive businesses with high growth potential.

Funding Stages

1. **Seed Stage:** Initial funding for idea development
2. **Series A:** First major round of venture capital financing
3. **Series B, C, etc.:** Subsequent rounds as the company grows

How VC Works

- The startup receives a **valuation**
- The VC receives an **equity stake** (e.g., 50%) in exchange for funding
- Funding is usually linked to specific **milestones** and business plan progress
- Investors can also be reached through **crowdfunding** websites

VC Investment Criteria

- High growth potential
- Scalable business model
- Strong management team
- Clear exit strategy (IPO or acquisition)

IX. Upcoming Topics

Remaining Material

- **Monetary Policy** - Handout to be provided
- **Market Microstructure** - Handout to be provided
- Most quantitative/numerical topics for the final exam have already been covered
- Read handouts before coming to class

X. Summary of Key Concepts

Key Takeaways

1. **Financing decisions** are flexible; **investment decisions** have finality
2. The **Pecking Order** is: Internal funds → Debt → New Equity
3. Debt cannot be unlimited; marginal cost increases with leverage
4. **Market values** (especially for equity) are preferred over book values for WACC
5. The 2008 crisis showed the danger of excessive leverage and complex securities
6. **Dual-class shares** allow founders to maintain control
7. **Venture capital** funds high-growth startups in exchange for equity

Formula Summary

$$\text{Market Value of Equity} = \text{Shares Outstanding} \times \text{Current Stock Price} \quad (1)$$

$$\text{Debt-to-Assets Ratio} = \frac{\text{Total Debt}}{\text{Total Assets}} \quad (2)$$

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Debt}}{\text{Total Equity}} \quad (3)$$

Key Terms Glossary

Term	Definition
Pecking Order	Financing priority: internal $\rightarrow$ debt $\rightarrow$ equity
Marginal Cost of Debt	The cost of each additional dollar of debt
MBS	Mortgage-Backed Securities (pools of mortgages)
CDO	Collateralized Debt Obligation (tranching debt securities)
Proxy	Authorization for someone to vote on behalf of a shareholder
Eurobond	Bond in a currency different from the issuance country
Panda Bond	Bond issued in China
Venture Capital	Equity capital for high-growth startups
Crowdfunding	Raising small amounts from many online investors